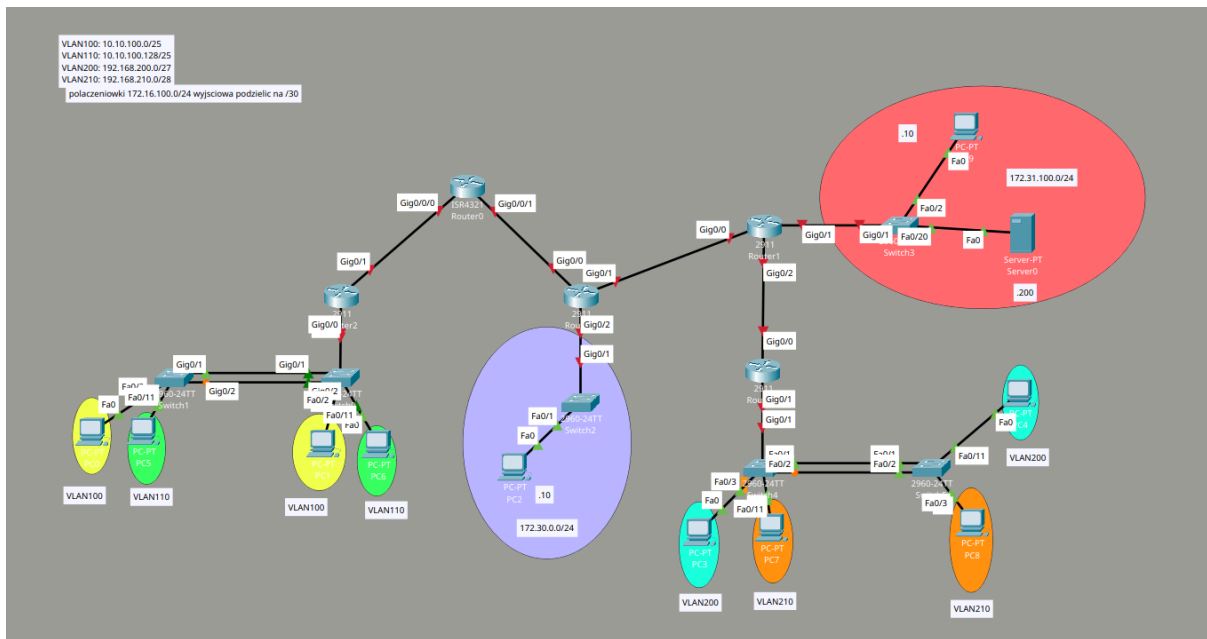


Cel:



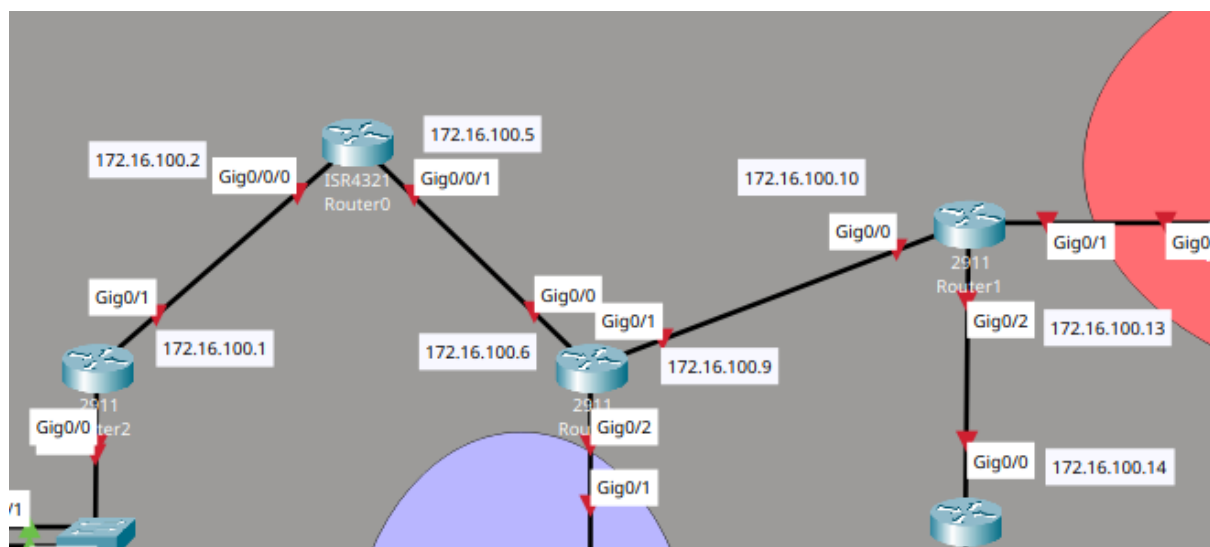
Rozwiązanie:

dzielenie sieci 172.16.100.0/24 na podsieci polaczeniowe

potrzebujemy 4 podsieci na połączenie routerów:

maska /30 (255.255.255.252)

1. NET: .0 FH: .1 LH:.2 BR: .3
2. .4 .5 .6 .7
3. .8 .9 .10 .11
4. .12 .13 .14 .15



konfiguracja tych podsieci na routerach:

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R2
R2(config)#int g0/0
R2(config-if)#ip add 172.16.100.1 255.255.255.252
R2(config-if)#no shut

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R2(config-if)#default int g0/0
Building configuration...

Interface GigabitEthernet0/0 set to default configuration
R2(config)#int g0/0
R2(config-if)#shut

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to down

R2(config-if)#int g0/1
R2(config-if)#ip add 172.16.100.1 255.255.255.252
R2(config-if)#no shut

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
|
```

(tu na początku zły interfejs)

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R0
R0(config)#int g0/0/0
R0(config-if)#ip add 172.16.100.2 255.255.255.252
R0(config-if)#no shut

R0(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

R0(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

R0(config-if)#int g0/0/1
R0(config-if)#ip add 172.16.100.5 255.255.255.252
R0(config-if)#no shut

R0(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

R0(config-if)#|
```

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R3
R3(config)#int g0/0
R3(config-if)#ip add 172.168.100.6 255.255.255.252
R3(config-if)#no shut

R3(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
int g0/1
R3(config-if)#ip add 172.16.100.9 255.255.255.252
R3(config-if)#no shut

R3(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

R3(config-if)#
```

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#int g0/0
R1(config-if)#ip add 172.16.100.10 255.255.255.252
R1(config-if)#no shut

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#int g0/2
R1(config-if)#ip add 172.16.100.13 255.255.255.252
R1(config-if)#no shut

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

R1(config-if)#|
```

```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#host R4
R4(config)#int g0/0
R4(config-if)#ip add 172.16.100.14 255.255.255.252
R4(config-if)#no shut

R4(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

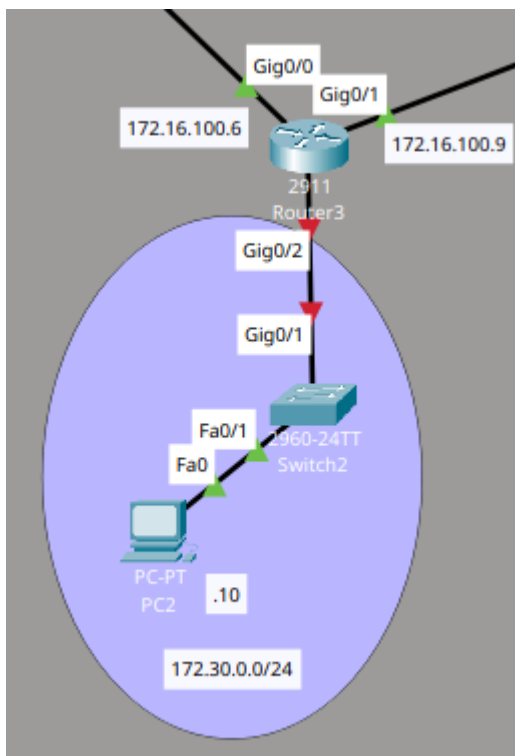
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R4(config-if)#|

```

teraz zajmujemy sie roznymi sieciami, tymi bez vlanow.

172.30.0.0/24



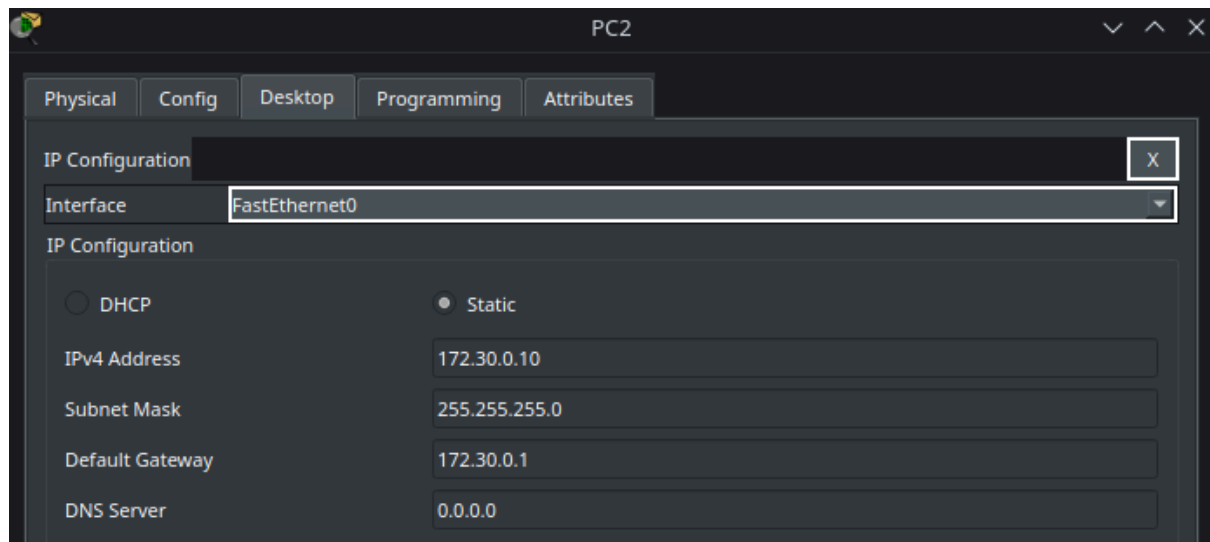
```

R3(config-if)#
R3(config-if)#int g0/2
R3(config-if)#ip add 172.30.0.1 255.255.255.0
R3(config-if)#no shut

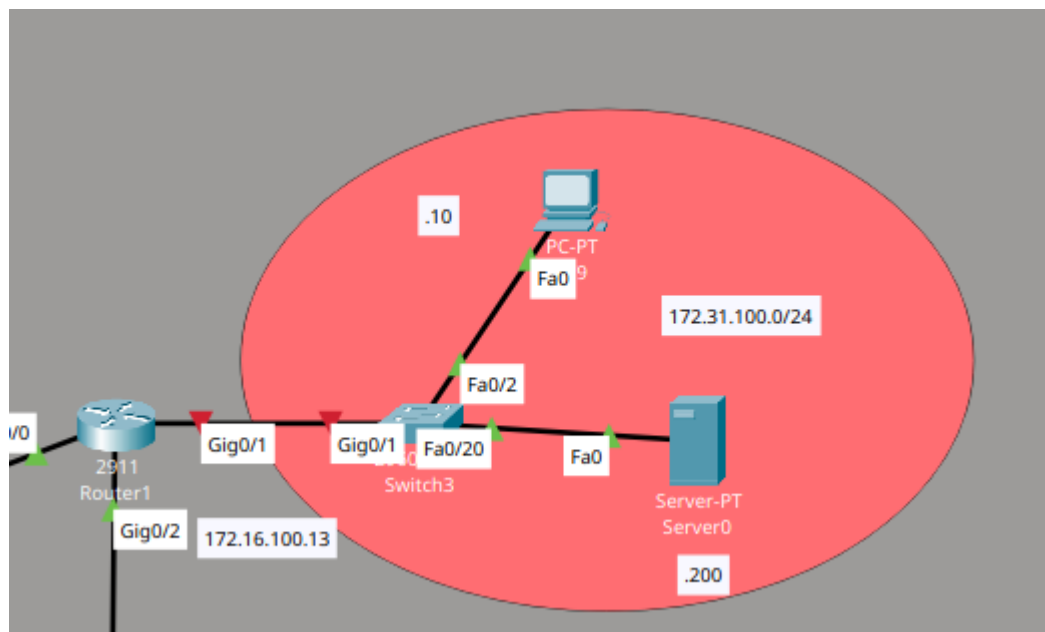
R3(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
|

```



172.31.100.0/24



```
R1(config-if)#int g0/1
R1(config-if)#ip add 172.31.100.1 255.255.255.0
R1(config-if)#no shut

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R1(config-if)#
```

PC9

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 172.31.100.10

Subnet Mask 255.255.255.0

Default Gateway 172.31.100.1

DNS Server 0.0.0.0

Server0

Physical Config Services Desktop Programming Attributes

IP Configuration X

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 172.31.100.200

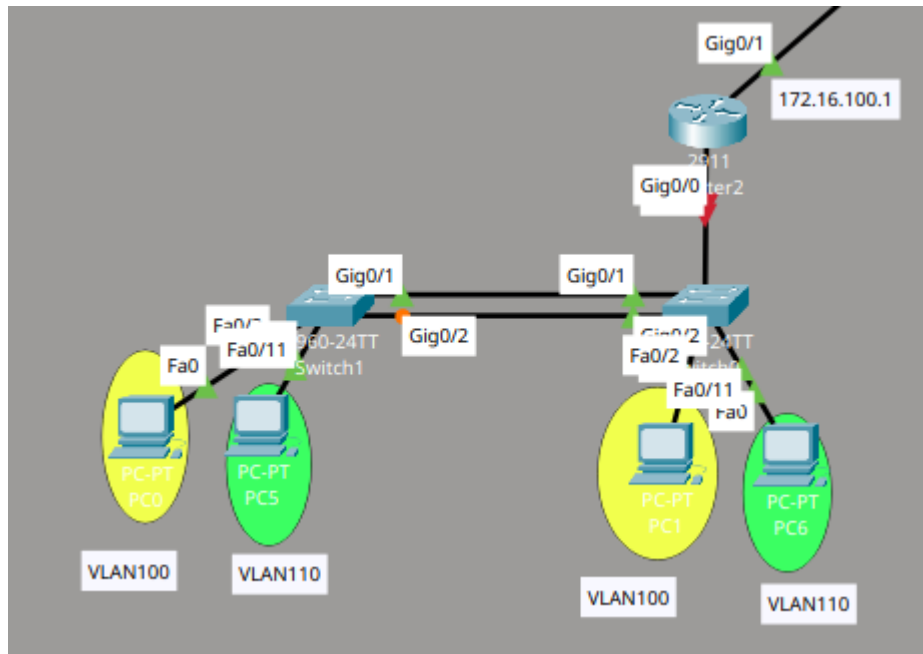
Subnet Mask 255.255.255.0

Default Gateway 172.31.100.1

DNS Server 0.0.0.0

teraz zabieramy się za vlany

zaczynamy od 100 i 110



najpierw stworzenie vlanow + lacp na switchach

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#host SW1
SW1(config)#vlan 100
SW1(config-vlan)#vlan 110
SW1(config-vlan)#int port-ch
SW1(config-vlan)#int port-channel 1
SW1(config-if)#int range g0/1-2
SW1(config-if-range)#sw m trunk

SW1(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

SW1(config-if-range)#sw tr al vlan 100,110
SW1(config-if-range)#channel
SW1(config-if-range)#channel-group 1 mode active
%EC-5-L3DONTBNDL2: Gig0/1 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Gig0/2 suspended: LACP currently not enabled on the remote port.
SW1(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

SW1(config-if-range)#

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#host SW0
SW0(config)#vlan 100
SW0(config-vlan)#vlan 110
SW0(config-vlan)#int port-channel 1
SW0(config-if)#int range gig0/1-2
SW0(config-if-range)#sw m trunk
SW0(config-if-range)#sw trunk all vlan 100,110
SW0(config-if-range)#ch
SW0(config-if-range)#channel
SW0(config-if-range)#channel-group 1 mode active
%EC-5-L3DONTBNDL2: Gig0/1 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Gig0/2 suspended: LACP currently not enabled on the remote port.
SW0(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface Port-channel1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

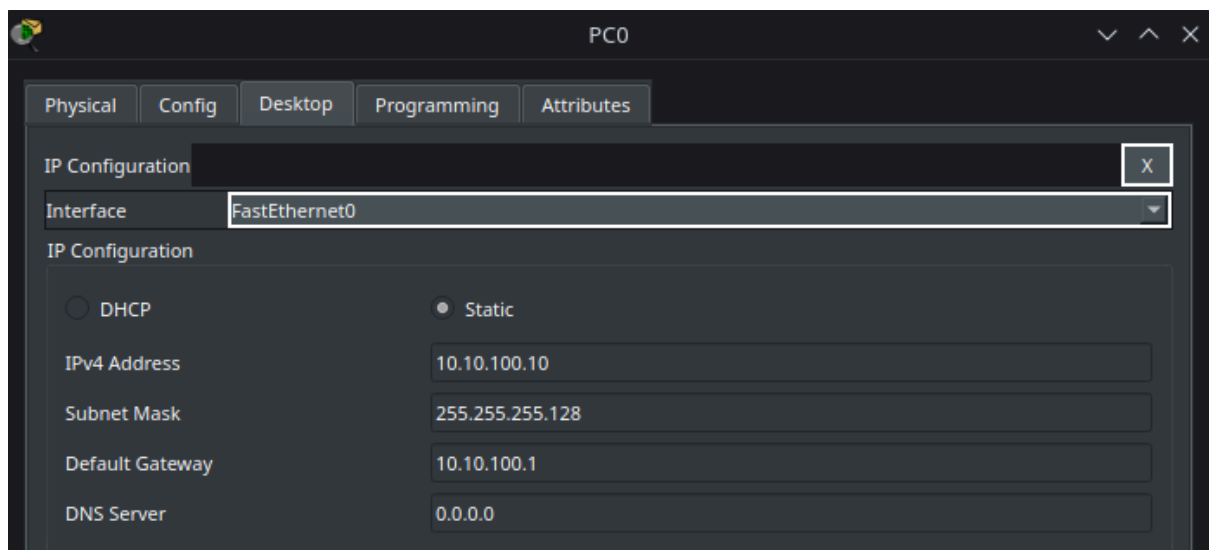
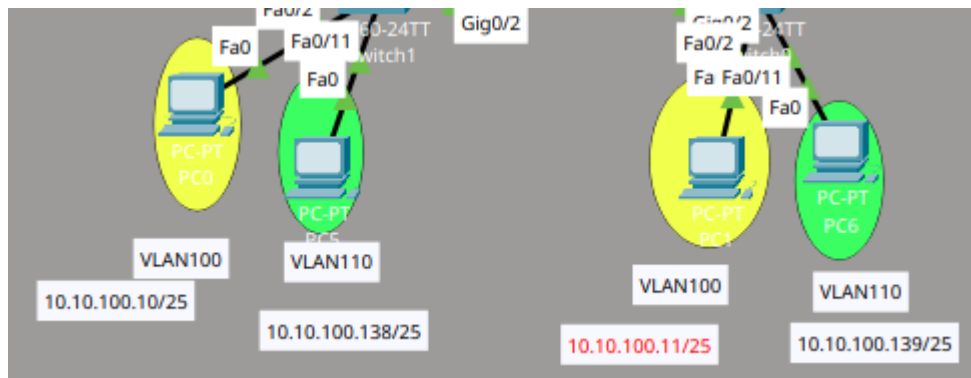
SW0(config-if-range)#
```

teraz przypisujemy vlany do interfejsow na switchach

```
SW1(config-if-range)#int f0/2
SW1(config-if)#sw m acc
SW1(config-if)#sw acc vlan 100
SW1(config-if)#int f0/11
SW1(config-if)#sw m acc
SW1(config-if)#sw acc vlan 110

SW0(config-if-range)#
SW0(config-if-range)#int f0/2
SW0(config-if)#sw m acc
SW0(config-if)#sw acc vlan 100
SW0(config-if)#int f0/11
SW0(config-if)#sw m acc
SW0(config-if)#sw acc vlan 110
```

i adresy do PCtow, ja wybralem takie:



PC5

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.10.100.138

Subnet Mask 255.255.255.128

Default Gateway 10.10.100.129

DNS Server 0.0.0.0

PC1

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.10.100.11

Subnet Mask 255.255.255.128

Default Gateway 10.10.100.1

DNS Server 0.0.0.0

PC6

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.10.100.139

Subnet Mask 255.255.255.128

Default Gateway 10.10.100.129

DNS Server 0.0.0.0

i w koncu konfiguracja ROAS:

```
R2>en
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int g0/0.100
R2(config-subif)#enca
R2(config-subif)#encapsulation do
R2(config-subif)#encapsulation dot1Q ?
<1-4094> IEEE 802.1Q VLAN ID
R2(config-subif)#encapsulation dot1Q 100 ?
native Make this as native vlan
<cr>
R2(config-subif)#encapsulation dot1Q 100
R2(config-subif)#no shut
R2(config-subif)#ip add 10.10.100.1 255.255.255.128
R2(config-subif)#int g0/0.110
R2(config-subif)#encapsulation dot1Q 110
R2(config-subif)#ip add 10.10.100.129 255.255.255.128
R2(config-subif)#no shut
R2(config-subif)#int g0/0
R2(config-if)#no shut

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0.100, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.100, changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0.110, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.110, changed state to up

R2(config-if)#
```

zapomnialem ustawic losowy native vlan i vlany na portchannel, wracam na swithe:

```
SW0(config-if)#
SW0(config-if)#int range g0/1-2
SW0(config-if-range)#sw m tr v ?
% Unrecognized command
SW0(config-if-range)#sw tr ?
allowed Set allowed VLAN characteristics when interface is in trunking mode
native Set trunking native characteristics when interface is in trunking
mode
SW0(config-if-range)#sw tr nat
SW0(config-if-range)#sw tr native vlan 500
SW0(config-if-range)#
```

```

SW0(config-if-range)#int port-channel 1
SW0(config-if)#sw m trunk
SW0(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINK-3-UPDOWN: Interface Port-channel1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface Port-channel1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

SW0(config-if)#sw tr al vlan 100,110
SW0(config-if)#sw tr native vlan 500
SW0(config-if)#

SW1>
SW1>en
SW1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#int range g0/1-2
SW1(config-if-range)#sw tr n
SW1(config-if-range)#sw tr native vlan 500
SW1(config-if-range)#

SW1(config-if-range)#int port-channel 1
SW1(config-if)#sw m trunk
SW1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINK-3-UPDOWN: Interface Port-channel1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface Port-channel1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

SW1(config-if)#sw trunk allowe
SW1(config-if)#sw trunk allowed vlan 100,110
SW1(config-if)#sw trunk native vlan 500
SW1(config-if)#

```

zapomnialem tez zrobic trunk do routera:

```

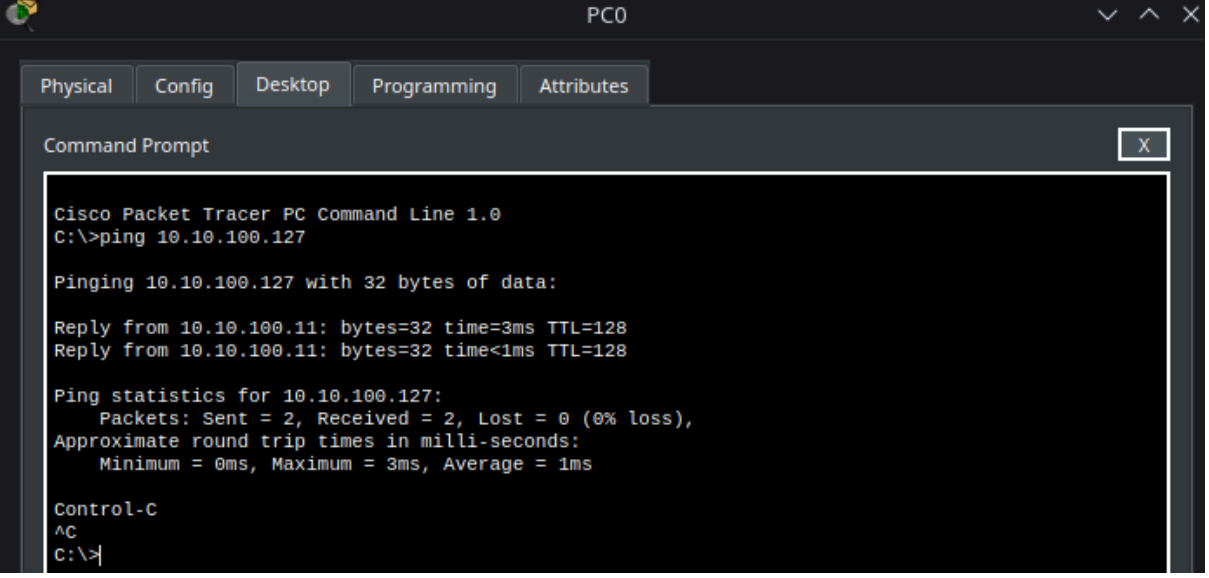
SW0(config-if)#sw tr al vlan 100,110
SW0(config-if)#sw tr native vlan 500
SW0(config-if)#int f0/1
SW0(config-if)#sw m trunk

SW0(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
sw t all vlan 100,110
SW0(config-if)#

```

test vlanow uzywajac adresow broadcastowych .128 i .255:



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.100.127

Pinging 10.10.100.127 with 32 bytes of data:

Reply from 10.10.100.11: bytes=32 time=3ms TTL=128
Reply from 10.10.100.11: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.100.127:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 1ms

Control-C
^C
C:\>|
```

odpowiada tylko .11, czyli dobry vlan, teraz ping na inny vlan specyficznie:

```
C:\>ping 10.10.100.139

Pinging 10.10.100.139 with 32 bytes of data:

Request timed out.
Reply from 10.10.100.139: bytes=32 time<1ms TTL=127
Reply from 10.10.100.139: bytes=32 time<1ms TTL=127
Reply from 10.10.100.139: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.100.139:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.10.100.138

Pinging 10.10.100.138 with 32 bytes of data:

Request timed out.
Reply from 10.10.100.138: bytes=32 time<1ms TTL=127
Reply from 10.10.100.138: bytes=32 time<1ms TTL=127
Reply from 10.10.100.138: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.100.138:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

ida, jest dobrze.

vlan 200 i 210

stworzenie vlanow, trunki i lacp na switchach:

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW4
SW4(config)#int port-channel 1
SW4(config-if)#exit
SW4(config)#vlan 200
SW4(config-vlan)#vlan 210
SW4(config-vlan)#int g0/1
SW4(config-if)#sw m trunk
SW4(config-if)#mw trunk all vlan 200,210
      ^
% Invalid input detected at '^' marker.

SW4(config-if)#sw trunk all vlan 200,210
SW4(config-if)#sw trunk nat vlan 500
SW4(config-if)#int range f0/1-2
SW4(config-if-range)#sw m trunk

SW4(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

SW4(config-if-range)#sw tr all vlan 200,210
SW4(config-if-range)#channel
SW4(config-if-range)#channel-gro
SW4(config-if-range)#channel-group 1 mode active
%EC-5-L3DONTBNDL2: Fa0/1 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Fa0/2 suspended: LACP currently not enabled on the remote port.
SW4(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

SW4(config-if-range)#int por
SW4(config-if-range)#int port-channel 1
SW4(config-if)#sw m trunk
SW4(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

SW4(config-if)#sw trunk vlan allo
SW4(config-if)#sw trunk al
SW4(config-if)#sw trunk allowed vlan
SW4(config-if)#sw trunk allowed vlan 200,210
SW4(config-if)#sw trunk native vlan 500
SW4(config-if)#
```

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW5
SW5(config)#vlan 200
SW5(config-vlan)#vlan 210
SW5(config-vlan)#int range fa0/1
SW5(config-if-range)#sw m trunk
SW5(config-if-range)#e
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with SW4 FastEthernet0/1 (500).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (1), with SW4 FastEthernet0/2 (500).exit
SW5(config)#int range fa0/1-2
SW5(config-if-range)#sw m trunk
SW5(config-if-range)#sw trunk allowed vlan 200,210
SW5(config-if-range)#channel-group 1 mode active
%EC-5-L3DONTBNDL2: Fa0/1 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Fa0/2 suspended: LACP currently not enabled on the remote port.
SW5(config-if-range)#
Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface Port-channel1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with SW4 FastEthernet0/1 (500).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (1), with SW4 FastEthernet0/1 (500).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with SW4 FastEthernet0/2 (500).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (1), with SW4 FastEthernet0/2 (500).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with SW4 Port-channel1 (500).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (1), with SW4 Port-channel1 (500).

SW5(config-if-range)#int port-channel 1
SW5(config-if)#sw m trunk
SW5(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINK-3-UPDOWN: Interface Port-channel1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface Port-channel1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with SW4 FastEthernet0/1 (500).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (1), with SW4 FastEthernet0/1 (500).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with SW4 FastEthernet0/2 (500).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (1), with SW4 FastEthernet0/2 (500).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with SW4 Port-channel1 (500).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (1), with SW4 Port-channel1 (500).

SW5(config-if)#sw trunk allowed vlan 200,210
SW5(config-if)#sw trunk native vlan 500
SW5(config-if)#
```

ROAS:

```
R4(config-if)#
R4(config-if)#int g0/1
R4(config-if)#no shut

R4(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R4(config-if)#int g0/1.200
R4(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/1.200, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.200, changed state to up

R4(config-subif)#encaps
R4(config-subif)#encapsulation dot1q 200
R4(config-subif)#ip add 192.168.200.1 255.255.255.224
R4(config-subif)#no shut
R4(config-subif)#int g0/1.210
R4(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/1.210, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.210, changed state to up

R4(config-subif)#encapsulation dot1q 210
R4(config-subif)#ip add 192.168.210.1 255.255.255.240
R4(config-subif)#no shut
R4(config-subif)#
```

dodawanie accessowych interfejsów i konfiguracja PCtów:

```
SW4(config-if)#int f0/3
SW4(config-if)#sw m acc
SW4(config-if)#sw acc vlan 200
SW4(config-if)#int f0.11
^
% Invalid input detected at '^' marker.

SW4(config-if)#int f0/11
SW4(config-if)#sw m acc
SW4(config-if)#sw acc 210
^
% Invalid input detected at '^' marker.

SW4(config-if)#sw acc vlan 210
SW4(config-if)#

SW5(config-if)#int f0/11
SW5(config-if)#sw m access
SW5(config-if)#sw m acc vlan 200
^
% Invalid input detected at '^' marker.

SW5(config-if)#sw acc vlan 200
SW5(config-if)#int f0/3
SW5(config-if)#sw m acc
SW5(config-if)#sw acc vlan 210
SW5(config-if)#
```

PC3

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.200.10

Subnet Mask 255.255.255.224

Default Gateway 192.168.200.1

DNS Server 0.0.0.0

PC7

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.210.10

Subnet Mask 255.255.255.240

Default Gateway 192.168.210.1

DNS Server 0.0.0.0

PC4

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.200.11

Subnet Mask 255.255.255.224

Default Gateway 192.168.200.1

DNS Server 0.0.0.0

PC8

Physical

Config

Desktop

Programming

Attributes

IP Configuration

Interface

FastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

192.168.210.11

Subnet Mask

255.255.255.240

Default Gateway

192.168.210.1

DNS Server

0.0.0.0

test vlanow uzywajac broadcastu i specyficznych adresow:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.210.15

Pinging 192.168.210.15 with 32 bytes of data:

Reply from 192.168.210.11: bytes=32 time<1ms TTL=128
Reply from 192.168.210.1: bytes=32 time<1ms TTL=255
Reply from 192.168.210.11: bytes=32 time<1ms TTL=128
Reply from 192.168.210.1: bytes=32 time<1ms TTL=255
Reply from 192.168.210.11: bytes=32 time<1ms TTL=128
Reply from 192.168.210.1: bytes=32 time<1ms TTL=255
Reply from 192.168.210.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.210.15:
    Packets: Sent = 4, Received = 7, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.200.10

Pinging 192.168.200.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.200.10: bytes=32 time<1ms TTL=127
Reply from 192.168.200.10: bytes=32 time=1ms TTL=127
Reply from 192.168.200.10: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.200.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.200.11

Pinging 192.168.200.11 with 32 bytes of data:

Request timed out.
Reply from 192.168.200.11: bytes=32 time<1ms TTL=127
Reply from 192.168.200.11: bytes=32 time=1ms TTL=127
Reply from 192.168.200.11: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.200.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>|
```

wszystko przechodzi, klasa

to czas w koncu na routing! jazda z ospfem

```
R2(config)#
R2(config)#router ospf 1
R2(config-router)#passive-interface g0/0
R2(config-router)#network 10.10.10.0 0.0.0.127 area 0
R2(config-router)#network 10.10.10.128 0.0.0.127 area 0
R2(config-router)#network 172.16.100.0 0.0.0.3 area 0
```

```

R0>en
R0#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R0(config)#router ospf 1
R0(config-router)#network 172.16.100.0 area 0
                                     ^
% Invalid input detected at '^' marker.

R0(config-router)#network 172.16.100.0 0.0.0.3 area 0
R0(config-router)#network 172.16.100.0 0.0.0.3 area 0
01:35:10: %OSPF-5-ADJCHG: Process 1, Nbr 172.16.100.1 on GigabitEthernet0/0/0 network
172.16.100.4 area 0
                                     ^
% Invalid input detected at '^' marker.

R0(config-router)#network 172.16.100.4 0.0.0.3 area 0
R0(config-router)#

```

```

R3>en
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)#passive-
R3(config-router)#passive-interface g0/2
R3(config-router)#network 172.16.100.4 0.0.0.3 area 0
R3(config-router)#network 172.16.100.8 0.0.0.3 area 0

R1>
R1>em
Translating "em"...domain server (255.255.255.255) % Name lookup aborted
R1>en
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#pas g0/1
R1(config-router)#network 172.16.100.8 0.0.0.3 area 0
R1(config-router)#network 172.16.100.12 0.0.0.3 area 0
R1(config-router)#|

R4(config-subif)#router ospf 1
R4(config-router)#passi
R4(config-router)#passive-interface g0/1
R4(config-router)#network 172.16.100.12 0.0.0.3 area 0
R4(config-router)#network
01:41:30: %OSPF-5-ADJCHG: Process 1, Nbr 172.31.100.1 on GigabitEthernet0/0 from LOADING to
FULL, Loadi\
                                     ^
% Invalid input detected at '^' marker.

R4(config-router)#network 192.168.200.0 0.0.0.31 area 0
R4(config-router)#network 192.168.210.0 0.0.0.15 area 0

```

zapomnialem o dwoch sieciach bez vlanow, wracam:

```

R3(config-router)#network 172.30.0.0 0.0.0.255 area 0
R3(config-router)#
R1(config-router)#network 172.31.100.0 0.0.0.255 area 0
R1(config-router)#

```

WIELKIE TESTOWANIE

wysylam pingi z PC0 na kazdy inny PC testujac lacznosc:

```
C:\>ping 10.10.100.138

Pinging 10.10.100.138 with 32 bytes of data:

Reply from 10.10.100.138: bytes=32 time=11ms TTL=127
Reply from 10.10.100.138: bytes=32 time=8ms TTL=127
Reply from 10.10.100.138: bytes=32 time<1ms TTL=127
Reply from 10.10.100.138: bytes=32 time=9ms TTL=127

Ping statistics for 10.10.100.138:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 7ms

C:\>ping 10.10.100.11

Pinging 10.10.100.11 with 32 bytes of data:

Reply from 10.10.100.11: bytes=32 time=12ms TTL=128
Reply from 10.10.100.11: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.100.11:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 6ms

Control-C
^C
C:\>ping 10.10.100.139

Pinging 10.10.100.139 with 32 bytes of data:

Reply from 10.10.100.139: bytes=32 time<1ms TTL=127
Reply from 10.10.100.139: bytes=32 time<1ms TTL=127
Reply from 10.10.100.139: bytes=32 time<1ms TTL=127
Reply from 10.10.100.139: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.100.139:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.30.0.10

Pinging 172.30.0.10 with 32 bytes of data:

Reply from 10.10.100.1: Destination host unreachable.
Reply from 10.10.100.1: Destination host unreachable.
Reply from 10.10.100.1: Destination host unreachable.
Reply from 10.10.100.1: Destination host unreachable.

Ping statistics for 172.30.0.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

ok, nie widzi kompa. sprawdzam tracert

```
C:\>tracert 172.30.0.10

Tracing route to 172.30.0.10 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      10.10.100.1
  2  0 ms      *          0 ms      10.10.100.1
  3  *          0 ms      *          Request timed out.
  4  0 ms      *          0 ms      10.10.100.1
  5  *          0 ms      *          Request timed out.
  6  0 ms      *          0 ms      10.10.100.1
  7  *          0 ms      *          Request timed out.
  8  0 ms      *          0 ms      10.10.100.1
  9  *          0 ms      *          Request timed out.
 10  0 ms      *          0 ms      10.10.100.1
 11  *          0 ms      *          Request timed out.
 12  0 ms      *          0 ms      10.10.100.1
 13  *          0 ms      *          Request timed out.
 14  0 ms      *          0 ms      10.10.100.1
 15  *          0 ms      *          Request timed out.
 16  0 ms      *          1 ms      10.10.100.1
 17  *          0 ms      *          Request timed out.
 18  0 ms      *          0 ms      10.10.100.1
 19  *          0 ms      *          Request timed out.
 20  0 ms      *          1 ms      10.10.100.1
 21  *          0 ms      *          Request timed out.
 22  0 ms
```

czyli nie wychodzimy poza router. sprawdzam czy R2 widzi R3 nie widzi.

```
R3(config-router)#do sh ip int br
Interface                IP-Address      OK? Method Status        Protocol
GigabitEthernet0/0       172.168.100.6   YES manual up             up
GigabitEthernet0/1       172.16.100.9    YES manual up             up
GigabitEthernet0/2       172.30.0.1      YES manual up             up
Vlan1                    unassigned      YES unset  administratively down down
```

R3 miał złe skonfigurowany interfejs, konfiguruje jeszcze raz

```
R3(config-router)#int g0/0
R3(config-if)#ip add 172.16.100.6 255.255.255.252
R3(config-if)#do sh ip int br
Interface                IP-Address      OK? Method Status        Protocol
GigabitEthernet0/0       172.16.100.6    YES manual up             up
GigabitEthernet0/1       172.16.100.9    YES manual up             up
GigabitEthernet0/2       172.30.0.1      YES manual up             up
Vlan1                    unassigned      YES unset  administratively down down
```

teraz robie ping z R2 jeszcze raz

```
R2#ping 172.16.100.6

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.100.6, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

działa, wracam do testowania polaczen z PC0

```
C:\>ping 172.30.0.10

Pinging 172.30.0.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.30.0.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

cały czas nie idzie, ale z R2 idzie:

```
R2#ping 172.30.0.10
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 172.30.0.10, timeout is 2 seconds:
```

```
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0
```

```
R0>en
```

```
R0#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
```

```
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
```

```
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
```

```
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
```

```
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
```

```
* - candidate default, U - per-user static route, o - ODR
```

```
P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```

    172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks
C       172.16.100.0/30 is directly connected, GigabitEthernet0/0/0
L       172.16.100.2/32 is directly connected, GigabitEthernet0/0/0
C       172.16.100.4/30 is directly connected, GigabitEthernet0/0/1
L       172.16.100.5/32 is directly connected, GigabitEthernet0/0/1
O       172.16.100.8/30 [110/2] via 172.16.100.6, 00:07:31, GigabitEthernet0/0/1
O       172.16.100.12/30 [110/3] via 172.16.100.6, 00:07:31, GigabitEthernet0/0/1
    172.30.0.0/24 is subnetted, 1 subnets
O       172.30.0.0/24 [110/2] via 172.16.100.6, 00:07:31, GigabitEthernet0/0/1
    172.31.0.0/24 is subnetted, 1 subnets
O       172.31.100.0/24 [110/3] via 172.16.100.6, 00:07:31, GigabitEthernet0/0/1
    192.168.200.0/27 is subnetted, 1 subnets
O       192.168.200.0/27 [110/4] via 172.16.100.6, 00:07:31, GigabitEthernet0/0/1
    192.168.210.0/28 is subnetted, 1 subnets
O       192.168.210.0/28 [110/4] via 172.16.100.6, 00:07:31, GigabitEthernet0/0/1
```

router 0 nie ma posieci vlanowych!

```
R2#show run | section ospf
```

```
router ospf 1
```

```
log-adjacency-changes
```

```
passive-interface GigabitEthernet0/0
```

```
network 10.10.10.0 0.0.0.127 area 0
```

```
network 10.10.10.128 0.0.0.127 area 0
```

```
network 172.16.100.0 0.0.0.3 area 0
```

```
R2#
```

blad znaleziony, zamiast 10.10.100... bylo 10.10.10 beka

naprawiam:

```
R2(config)#router ospf 1
```

```
R2(config-router)#no network 10.10.10.0
```

```
% Incomplete command.
```

```
R2(config-router)#no network 10.10.10.0 0.0.0.127 area 0
```

```
R2(config-router)#no network 10.10.10.128 0.0.0.127 area 0
```

```
R2(config-router)#network 10.10.100.128 0.0.0.127 area 0
```

```
R2(config-router)#network 10.10.100.0 0.0.0.127 area 0
```

```
R2(config-router)#do show run | section ospf
```

```
router ospf 1
```

```
log-adjacency-changes
```

```
passive-interface GigabitEthernet0/0
```

```
network 172.16.100.0 0.0.0.3 area 0
```

```
network 10.10.100.128 0.0.0.127 area 0
```

```
network 10.10.100.0 0.0.0.127 area 0
```

```
R2(config-router)#
```

i w koncu testujemy dalej!

```
C:\>ping 172.30.0.10

Pinging 172.30.0.10 with 32 bytes of data:

Reply from 172.30.0.10: bytes=32 time<1ms TTL=125
Reply from 172.30.0.10: bytes=32 time<1ms TTL=125
Reply from 172.30.0.10: bytes=32 time<1ms TTL=125
Reply from 172.30.0.10: bytes=32 time=5ms TTL=125

Ping statistics for 172.30.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 5ms, Average = 1ms

C:\>ping 172.31.100.10

Pinging 172.31.100.10 with 32 bytes of data:

Request timed out.
Reply from 172.31.100.10: bytes=32 time<1ms TTL=124
Reply from 172.31.100.10: bytes=32 time<1ms TTL=124
Reply from 172.31.100.10: bytes=32 time<1ms TTL=124

Ping statistics for 172.31.100.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.31.100.200

Pinging 172.31.100.200 with 32 bytes of data:

Request timed out.
Reply from 172.31.100.200: bytes=32 time=10ms TTL=124
Reply from 172.31.100.200: bytes=32 time<1ms TTL=124
Reply from 172.31.100.200: bytes=32 time<1ms TTL=124

Ping statistics for 172.31.100.200:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>ping 192.168.200.10

Pinging 192.168.200.10 with 32 bytes of data:

Reply from 192.168.200.10: bytes=32 time<1ms TTL=123
Reply from 192.168.200.10: bytes=32 time<1ms TTL=123
Reply from 192.168.200.10: bytes=32 time=1ms TTL=123
Reply from 192.168.200.10: bytes=32 time<1ms TTL=123

Ping statistics for 192.168.200.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
C:\>ping 192.168.200.11

Pinging 192.168.200.11 with 32 bytes of data:

Reply from 192.168.200.11: bytes=32 time<1ms TTL=123
Reply from 192.168.200.11: bytes=32 time<1ms TTL=123
Reply from 192.168.200.11: bytes=32 time=10ms TTL=123
Reply from 192.168.200.11: bytes=32 time<1ms TTL=123

Ping statistics for 192.168.200.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ping 192.168.210.10

Pinging 192.168.210.10 with 32 bytes of data:

Reply from 192.168.210.10: bytes=32 time<1ms TTL=123
Reply from 192.168.210.10: bytes=32 time=1ms TTL=123
Reply from 192.168.210.10: bytes=32 time=2ms TTL=123
Reply from 192.168.210.10: bytes=32 time<1ms TTL=123

Ping statistics for 192.168.210.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 192.168.210.11

Pinging 192.168.210.11 with 32 bytes of data:

Reply from 192.168.210.11: bytes=32 time<1ms TTL=123
Reply from 192.168.210.11: bytes=32 time<1ms TTL=123
Reply from 192.168.210.11: bytes=32 time<1ms TTL=123
Reply from 192.168.210.11: bytes=32 time<1ms TTL=123

Ping statistics for 192.168.210.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

wszystko sie widzi! koniec laba